AICHU TAN

Dublin, CA 94568 | the.aichutan@gmail.com | Portfolio: https://AichuTan.github.io |
LinkedIn: https://www.linkedin.com/in/aichutan/

Professional Summary

Data analyst and M.S. in Big Data Analytics candidate (expected 2026) with experience designing full-stack analytical solutions across agriculture, urban equity, and public health. Skilled in Python, R, SQL, Tableau, and cloud platforms. Passionate about creating data tools that drive social and organizational impact.

Technical Skills

Programming & Scripting: Python, R, SQL, JavaScript, C++
Data Science & ML: YOLOv8, LangChain, TensorFlow, Scikit-learn, RAG
Data Analysis & Visualization: Pandas, NumPy, ggplot2, Tableau, ArcGIS
Cloud & Infrastructure: AWS, Google Cloud, Azure, BigQuery, RDS
Web & Tools: HTML/CSS, Streamlit, GitHub Pages, Jekyll, Git

Project Experience

AI-Powered Plant Disease Detection
Tools: Python, YOLOv8, LangChain, LLM
Designed an AI system to help farmers in low-resource settings identify crop diseases and receive immediate treatment guidance.

- Built YOLOv8 model for real-time disease detection from plant images
- Integrated LangChain + LLMs to generate treatment and prevention advice
- Deployed web-based app to support precision agriculture in underserved communities

Mapping for Change
Tools: HTML, ArcGIS Dashboard, GitHub Pages

- Created an interactive web platform visualizing homelessness, rent burden, and tree equity in Downtown San Diego.
- Leveraged ArcGIS and public datasets to support policy advocacy for underserved communities.

Online Ticket Database Management System
Tools: MySQL, AWS RDS, ERD, Python, Datapane

- Designed and implemented a relational database and schema for an event ticketing system.
- Built a Python-based dashboard to analyze revenue, booking trends, and customer insights.

Bellabeat Data Analysis Case Study
Tools: R, tidyverse, ggplot2

- Analyzed smart device data to uncover behavioral trends in sleep, activity, and calorie burn.
- Delivered marketing strategy recommendations to optimize user engagement and retention.

Professional Development

- **LangChain for LLM Application Development** – Coursera (Harrison Chase, Andrew Ng)
- **Streamlit for Data Science** – Udemy (Marco Peixeiro)
- **100 Days of Code: Python Pro Bootcamp** – Udemy (Dr. Angela Yu)
- **Full-Stack Web Development Bootcamp** – Udemy (Dr. Angela Yu)
- **YOLOv7: Deep Learning for Object Detection** – Udemy (Dr. Priyanto Hidayatullah)
- **AI A–Z: Agentic AI, GenAI & Reinforcement Learning** – Udemy (Hadelin de Ponteves)
- **Machine Learning A–Z (Python & R)** – Udemy (Kirill Eremenko, Hadelin de Ponteves)

Education

San Diego State University (SDSU), San Diego, CA
Master of Science in Big Data Analytics | GPA: 4.0 | Aug 2024 – Present

National Cheng Kung University (NCKU), Taiwan
Bachelor of Science in Chemical Engineering

Google Data Analytics Professional Certificate — Coursera
Completed Dec 2022

Work Experience

Business Owner – Wholesale Clothing
Taiwan | Apr 2009 – Oct 2016

- Increased revenue by 50% through targeted market research and pricing strategy.
- Managed supply chain logistics, vendor negotiations, and inventory forecasting.
- Oversaw import/export operations across Taiwan, China, and Southeast Asia.

Process Engineer – ProMOS Technology (DRAM)
Taiwan | Jul 2004 – Feb 2009

- Implemented Statistical Process Control (SPC) to reduce yield loss and manufacturing defects.
- Conducted root cause and variance analyses to optimize semiconductor production processes.

Assistant Research Professor – NCKU
Taiwan | Jul 2003 – Jun 2004
- Researched nanoparticle-biomolecule interactions for medical applications

Publication

Tan, A., Vito, D., & Fernandez, G. (2025). *AI for Farmers: A YOLO and Language Model-Based Plant Disease Detection and Advisory System*. Presented at the 4th International One Health Conference.

Lin, H.-Y., Lay, E., Wen, W.-Y., Dewi, H., Cheng, Y.-C., Tsai, S.-W., & Tan, A. (2004). *Racemization and hydrolysis of (S)-naproxen 2,2,2-trifluoroethyl ester in non-polar solvents. Journal of Physical Organic Chemistry*. https://doi.org/10.1002/poc.810 Cited by 6 (as of 2025)